

SIMATS ENGINEERING

ASSESSMENT TOOL 3 - UNIT 4

Capstone Project

COURSE NAME:Principles of Digital System Design
COURSE CODE:ECA47

COURSE OUTCOME COVERED

CO4: Apply and evaluate microcontroller (8051/Arduino) architecture, instruction sets, and interfacing techniques to develop embedded system applications. (BL6)

Assessment Tool Weightage: 45%

Duration: 3 hours

Marks:100

This is a comprehensive project-based assessment carried out as an individual or team activity, where students are assigned a real-world problem. The project should demonstrate problem-solving ability, application of engineering principles, and system-level thinking. Evaluation is based on innovation, design effectiveness, integration of concepts, performance optimization, and clarity in report and presentation.

TOPICS: Introduction to Microcontroller - Pin configuration - Architecture of 8051 - Instruction sets - Addressing modes-I/O Ports-External memory-Timers and Counters-Serial data-input and output-Interrupts- Interfacing with ADC, DAC, LEDs and seven segment display. Introduction to Arduino - Arduino IDE - Creating an Arduino program - Analog and Digital Interfacing - Adding Interrupts - communicating with devices and sensors - IOT application- Alphanumeric LCD display - Temperature and humidity sensor interfacing

Criterion	Weightage (%)	Maximum Marks	Marks Awarded
Conceptual Understanding	25%	25	_____
Application	25%	25	_____
Analytical Thinking	20%	20	_____
Communication	15%	15	_____
Response to Questions	15%	15	_____
Total	100%	100	_____ / 100

QUESTIONS: [any 1 for a group of 4]

1 x 100 = 100

1. Development of an Obstacle-Avoiding Robot Using Arduino and Ultrasonic Sensors. BL6 (100)
2. Audio Amplifier Circuit Design and Simulation. BL6 (100)
3. Alcohol Detector using Arduino. BL4 (100)
4. AquaSentinel: Intelligent Water Level Monitoring System. BL6 (100)
5. Design and Implementation of an Electronic Night Lamp. BL4 (100)
6. Design and Implementation of a Low Cost Radar System Using Arduino for Object Detection. BL6 (100)
7. An electronic election system to securely register and count votes. BL6
8. Laser-Based Security Alarm System. BL3 (100)
9. "Building a Traffic Signal Controller using Flip-Flops, Counters, and Shift Registers". BL6 (100)
10. Smart Classroom Energy Server using Logic Gates and Sensors. BL6
11. Temperature-Based Fan Controller. BL4 (100)
12. Arduino-based smart voice-controlled digital lock system. BL6 (100)
13. Smart Automatic Night Light: A Sensor - Based Energy Saving Solution. BL4 (100)
14. LED Flasher Using 555 Timer (Astable Mode). BL4 (100)
15. Smart IR doorbell sensor for Motion detection and security. BL4 (100)
16. Door Alarm System Using Push Button and Buzzer. BL4 (100)
17. Wireless Power Transfer System. BL4 (100)
18. Design and Implementation of a Low Cost Radar System Using Arduino for Object Detection. BL6 (100)
19. Fire Alarm System. BL4 (100)
20. Smart Fire Detection and Alarm System. BL6 (100)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25%	Explains concepts accurately with complete clarity; demonstrates deep understanding	Explains concepts correctly with minor gaps	Partial understanding; explanations contain errors or omissions	Unable to explain basic concepts
Application	25%	Applies concepts correctly to new situations; solves problems logically and independently	Applies concepts to familiar problems with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20%	Analyzes scenarios critically; provides strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis; weak justification	No analytical reasoning demonstrated
Communication	15%	Communicates ideas clearly, logically, and confidently with proper terminology	Communicates clearly with minor hesitation or lack of precision	Communication is unclear or lacks structure	Unable to communicate ideas effectively
Response to Questions	15%	Responds accurately, confidently, and extends answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions